The Crossing Number of Complete Bipartite Graphs

M309: Step-Level Analysis, Ram Reformulation,
Exact Rank Certificates, and Local Extensions

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Abstract

This report examines the September 23, 2026 manuscript at the exact catalogue pin, together with the complete private report `TR-item-071-astra.tex`. The ordinary point-count crossing problem, one common drawing, signed endpoint corrections, an interpolation detector, an arbitrary-subspace inequality, and the odd/even deletion argument are treated separately. We give complete finite-algebra proofs and an explicit conditional reconstruction of the manuscript's global deduction, keeping its geometric normalization and cycle-pairing contract visible. Ram representations preserve these finite spaces and constraints; they do not manufacture drawing realizability.

Local contributions proved here include an exact nonnegative decomposition of the subspace-bound defect, equality restrictions on extremal drawing certificates, integer-weighted rank bounds, a coordinate-invariant detector certificate, and the ceiling-sensitive odd correction in the rotation-frequency relaxation. Rational common drawings, detector orders, and subspace filtrations are checked with exact arithmetic in two independent implementations. Ten source files are byte-matched to the pin. Lean solution and challenge artifacts are available; their full dependency chain and comparator have not been replayed in this audit. Finite tests are not certification of all drawings, and no independent priority claim for the global formula is made.

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1 Scope, Sources, and Claim Status

1.1 Version and reading contract

Catalogue row 309 identifies *The crossing number of complete bipartite graphs*, directory

`preprints/The-crossing-number-of-complete-bipartite-graphs-September-23-2026`

and commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. The local `build` directory contains `paper.tex`, `references.bib`, six section files, and two figure sources. All ten were read and byte-matched to their public counterparts at that commit. The local source is an extracted snapshot, not a Git worktree; the pin assertion rests on this comparison and the catalogue, not a local Git HEAD. Paths and reading depth are in Appendix A.

The fully read private crossing report [2] is dated September 6, 2026. Its open-problem statement describes that baseline; it is not a refutation of the later pinned manuscript. Conversely, the manuscript's proof claim does not turn the private report's conditional copositive targets into established inequalities. We distinguish:

1. **Local derivations proved here:** finite linear algebra, coordinate-transfer, counting, and the explicitly conditional implications with full proofs below.

2. **Source geometry:** normalization and cycle pairing for every admissible continuous-arc drawing. The construction was fully read and reconstructed; an independent local manual check also covered the full normalization and intersection sections and found no concrete defect. This is a positive scoped manual check, not an all-input machine replay.

3. **Formalization artifacts:** a named Lean solution theorem, challenge definitions, and comparator configuration. Their presence is recorded accurately in Section 10.3; no replay is claimed.

The independent algebra audit found no concrete error in the detector, subspace argument, signed-to-unsigned conversion, or parity reduction. We do not assert a counterexample or identify an unproved algebra step. The global theorem is nevertheless displayed conditionally on its geometric interface, keeping the scope of our bounded verification separate from the source's universal claim.

1.2 Public refresh and attribution

Public searches on October 7 used public manuscript and article titles only, never private corpus titles or contents. The primary Brosch–Polak paper [3] gives rotation-frequency and symmetry-reduced semidefinite bounds, including $\mathrm{cr}(K_{10,n}) \geq 4.87057n^2 - 10n$ and $\mathrm{cr}(K_{13,n}) \geq 8.65675n^2 - 18n$. Its published volume is **207** (2024), pages 693–715, correcting the private report's metadata; the author's record [4] corroborates this. The publication by Balogh, Lidický, Norin, Pfender, Salazar, and Spiro [5] gives the balanced asymptotic coefficient 0.9118. These are comparison baselines, not independent verification of the later global proof.

Lagrange interpolation, determinant multiplicities, graph-of-map intersections, signed intersections, convex averaging, copositivity, and deletion double counting are classical or elementary mechanisms. The particular detector and subspace-filtration argument below are attributed to M309 [1], not to a pre-existing Ram theorem. The Ram contribution is an exact representation and verification architecture, plus the explicitly derived local refinements.

2 The Problem and Its Quantifiers

For positive integers m, n, write

$$h(r) = \left\lfloor \frac{r}{2} \right\rfloor \left\lfloor \frac{r-1}{2} \right\rfloor, \qquad Z(m,n) = h(m)h(n).$$

The graph parts are $A = \{a_1, \ldots, a_m\}$ and $B = \{b_1, \ldots, b_n\}$. A drawing places vertices injectively in the plane and uses simple continuous edge arcs whose interiors avoid vertices. Distinct interiors meet at finitely many proper crossings, without triple points, tangencies, or common subarcs. Its count $c(D)$ counts every crossing point, including repeated crossings of a pair. Thus $\mathrm{cr}(K_{m,n}) = \min_{D \in \mathcal{D}_{m,n}} c(D)$, and the claimed formula means

$$\forall m, n \geq 1: \quad [\exists D : c(D) = Z(m,n)] \quad \text{and} \quad [\forall D : c(D) \geq Z(m,n)], \qquad D \in \mathcal{D}_{m,n}. \tag{2.1}$$

This is not the rectilinear, two-page, pair-crossing, odd-crossing, or forbidden-biclique extremal problem. The distinctions in [8, 9] are material.

2.1 The upper bound

Split each part evenly between positive and negative rays of its axis, use the vertical axis for A and horizontal axis for B, and draw straight segments. In a quadrant with u, v vertices on its bounding rays, each choice of two on each ray forms a convex quadrilateral whose diagonals are the unique crossing pair. Its contribution is $\binom{u}{2}\binom{v}{2}$. Summing gives $h(m)h(n)$, since

$$\binom{\lfloor r/2 \rfloor}{2} + \binom{\lceil r/2 \rceil}{2} = h(r).$$

Triple concurrence is avoided by choosing intercepts off the finitely many concurrence polynomial zero sets. Each polynomial is nonzero: fix two independent segments and vary an intercept of the third. A finite product of nonzero real polynomials cannot vanish on an open parameter box. Adjacent segments cannot meet again in their interiors. This proves the upper bound. If one part has size at most two, nonnegativity already proves equality.

2.2 Normalization and the geometric contract

Source `lem:normalization` reduces any admissible drawing, without increasing $c(D)$, to polygonal edges, straight near vertices and crossings, without adjacent-edge crossings. Independent edges may still cross more than once. Its four stages are:

1. Choose disjoint round disks around the finitely many vertices and crossings. On each incident arc remove the interval from first to last visit to its closed disk. Compactness separates all other pieces. This does not assume finitely many visits to a disk boundary.

2. The retained compact arcs have disjoint open neighborhoods. Short exterior radial spurs can be connected through exterior collars; polygonal paths in those open neighborhoods join the spurs. Deleting loops in the resulting finite embedded graph makes each arc simple.

3. Fill crossing disks by two chords, with at most one crossing, and vertex disks by spokes from generic interior points. Disjoint neighborhoods prevent other new intersections.

4. At an adjacent-edge crossing exchange prefixes from the common vertex and reconnect by disjoint chords. Every exterior track is used once in total. Delete self-loops in the two new walks. Retained third-edge crossings remain straight-through; a newly created turn is isolated from other edges. The crossing count strictly decreases, so iteration terminates.

These points are necessary for ordinary point counting. Existence of a minimum follows because a nonempty set of finite nonnegative integer counts has a least member, not from a compactness theorem on drawings.

Definition 2.1 (Geometric contract G)**.** Every admissible drawing can be normalized as above. For every normalized drawing, the matrix J of Section 3, formed from one common drawing and one set of vertex cuts, satisfies $z^T J y = 0$ for all real graph-cycle vectors z, y.

G is the geometric interface of the conditional global argument. It does not claim that arbitrary prescribed rotations or signed matrices are realizable. The source gives a construction intended to prove G; we reconstruct that construction next.

3 Signed Intersections and Common-Drawing Compatibility

Orient ip from a_i to b_p. Let $I(e, f)$ be the signed sum of interior crossings, positive when the ordered edge directions give the positive orientation of the plane. At a vertex v fix a gap and list outward spokes counterclockwise. Set $T_v(e, f) = 1$ if e precedes f, -1 if it follows, and zero if $e = f$. Put $\sigma_{ve} = 1$ for an outgoing edge and -1 for an incoming edge. Define the skew-symmetric matrix

$$J(e, f) = I(e, f) + \frac{1}{2} \sum_{v \text{ common endpoint}} \sigma_{ve}\sigma_{vf} T_v(e, f), \qquad J^{ik}(p, q) = J(ip, kq). \tag{3.1}$$

For $i \neq k$, $p \neq q$ the entry is exactly $I(ip, kq)$. The diagonal block J^{ii} is half a linear spoke-order sign matrix because adjacent edges have no interior crossings.

Lemma 3.1 (Local chord signs)**.** *For oriented chords C, E with four distinct boundary endpoints, assign coefficient -1 to a start and $+1$ to an end. With T the counterclockwise comparison from a boundary cut,*

$$I(C,E) = \frac{1}{2} \sum_{u \in \partial C} \sum_{w \in \partial E} \varepsilon_C(u) \varepsilon_E(w) T(u,w).$$

Proof. Passing the cut through one endpoint changes the expression by a multiple of the other chord's coefficient sum, which is zero. Cut before the start of C. The expression is minus the sum of E's coefficients on the counterclockwise arc from the start to the end of C, lying to the right of that directed chord. Nonalternating endpoints give zero. Alternating endpoints give $+1$ for a right-to-left crossing by E and -1 otherwise, the required determinant signs. □

3.1 Reconstruction of the source cycle argument

Replace the first cycle in every vertex disk by its joining chord, keeping exterior edge tracks. For the second cycle make two walks: shift every edge track slightly left and slightly right relative to the fixed graph-edge orientation, not each cycle traversal. In crossing disks, the shifted track meets the other original track once with the original sign and avoids its own original track. In a bend disk the two shifted endpoints lie on the same side of the original bent arc and can be joined there. Other truncated compact pieces are disjoint and positively separated; finiteness allows globally small shifts. Vertex-disk chords join the displaced endpoints, with no seam or corner contacts.

The chord endpoint coefficient on spoke e is $\sigma_{ve} z_e$ for the first cycle and $\sigma_{ve} y_e$ for the second. Different spokes retain their comparison signs in both shifted versions. On a shared spoke the two displaced endpoints are on opposite sides of the original in the boundary order, so their signs average to zero. Lemma 3.1 gives the averaged vertex contribution

$$\frac{1}{2} \sum_{e,f \ni v} z_e y_f \sigma_{ve} \sigma_{vf} T_v(e,f).$$

The exterior contribution is $z^T I y$ in each version. Opposite traversals of a shared edge change coefficients, not this cancellation.

Two transverse closed polygonal walks have algebraic intersection zero even if they self-intersect. Cone each directed segment $[u,w]$ of the first to a generic point o. Each triangle boundary has zero net intersection with the second closed walk, since entrances and exits have opposite signs. Choose o off the finitely many lines causing degeneracy or corner contacts. Summing the boundaries cancels the cone segments and leaves the first walk, counted with traversal multiplicity. Applying this to both shifted versions and averaging gives exactly $z^T J y = 0$.

This is the source's geometric proof, not an inference from numerical matrices. The universal construction must avoid tracks in every bend disk and retain transversality at seams, including common edges. The independent local manual check of the complete sections found no concrete defect in these steps. Exact examples below support the signs but do not themselves establish the universal quantifier.

3.2 The compatibility quotient

Let $\mathcal{L} = \{H : H(p,q) = r_p + c_q,\ r, c \in \mathbb{R}^n\}$.

Proposition 3.2. *If J vanishes on pairs of graph cycles, then for all i, k,*

$$J^{ik} - J^{i1} - J^{1k} + J^{11} \in \mathcal{L}. \tag{3.2}$$

Proof. For $i,k \neq 1$ use the four-cycle with entries $+1$ on $ip, 1p'$ and -1 on $ip', 1p$, and the corresponding cycle for k, q, q'. Expanding the zero pairing gives, for the matrix H in (3.2), $H(p,q) - H(p,q') - H(p',q) + H(p',q') = 0$. Repeated indices are trivial. Fixing p_0, q_0 gives $H(p,q) = H(p,q_0) + H(p_0,q) - H(p_0,q_0)$, a row-plus-column form. If $i = 1$ or $k = 1$, $H = 0$ directly. □

The four-cycles through one fixed vertex in each part form a basis: each other edge has its own unique coordinate, and the cycle dimension is $mn - (m+n) + 1 = (m-1)(n-1)$. Thus $C^T J C = 0$ for their column matrix is an exact finite pairing test. Passing it is not sufficient for arbitrary data to be realizable by a plane drawing.

4 The Subspace Bound, Reproved

Theorem 4.1 (M309 subspace inequality)**.** *For $U_i, V_i \subseteq W$, $1 \leq i \leq m$, with $\dim W = N$ and $U_i \cap V_i = 0$,*

$$\sum_{i,j} \dim(U_i \cap V_j) \leq C_m N, \qquad C_m = \lfloor m^2/4 \rfloor. \tag{4.1}$$

Proof. Choose distinct real nodes $t_1, \ldots, t_m$. For $Z \subseteq W$ define

$$\mathcal{P}_h(Z) = \{P \in Z[\xi] : \deg P \leq h,\ P(t_i) \in U_i\ \forall i\}, \qquad \mathcal{P}_{-1}(Z) = 0,$$
$$k_h(Z) = \dim \mathcal{P}_h(Z) - \dim \mathcal{P}_{h-1}(Z).$$

At a non-node t, the kernel of evaluation is $(\xi - t)\mathcal{P}_{h-1}(Z)$: division preserves coefficients in Z and all constraints because $t_i - t \neq 0$. Hence, writing $E_h(Z;t) = \{P(t) : P \in \mathcal{P}_h(Z)\}$,

$$\dim E_h(Z;t) = k_h(Z) \quad (t \notin \{t_i\}). \tag{4.2}$$

Lagrange interpolation is an isomorphism $\mathcal{P}_{m-1}(Z) \to \bigoplus_i (U_i \cap Z)$ with inverse $(z_i) \mapsto \sum_i L_i(\xi) z_i$. Uniqueness follows from m roots of a polynomial of degree at most $m-1$, coordinatewise. Therefore

$$\sum_{h=0}^{m-1} k_h(Z) = \sum_i \dim(U_i \cap Z). \tag{4.3}$$

Fix a non-node t_*. Nested spaces $E_h(W;t_*)$ imply $\ell_h = k_h(W) - k_{h-1}(W) \geq 0$, where $k_{-1} = 0$. Choose ℓ_h polynomials of degree at most h at stage h whose values extend the previously chosen basis. The matrix M_h of all columns chosen through h has $k_h(W)$ independent columns at t_*. Choose a square row minor Δ_h nonzero there. Its column degrees give

$$\deg \Delta_h \leq \sum_{e=0}^{h} e\ell_e. \tag{4.4}$$

If there are no columns use the empty determinant 1.

Let $\pi_j : W \to W/V_j$. For every non-node t, the kernel of π_j on $E_h(W;t)$ contains $E_h(V_j;t)$; its projected rank is at most $k_h(W) - k_h(V_j)$. Express evaluation of a fixed basis of $\mathcal{P}_h(W)$ followed by π_j as a polynomial matrix. Its larger minors vanish at infinitely many non-nodes, hence identically, including at t_j. Since $E_h(W;t_j) \subseteq U_j$ and π_j is injective on U_j, this implies $\dim E_h(W;t_j) \leq k_h(W) - k_h(V_j)$. Thus Δ_h has rank deficiency at least $k_h(V_j)$ at t_j. A square polynomial matrix of deficiency q at a node has determinant divisible by $(\xi - t_j)^q$: put q constant

kernel vectors into an invertible column change and factor $\xi - t_j$ from those columns. Distinct-node factors multiply. Consequently

$$\sum_j k_h(V_j) \le \deg \Delta_h \le \sum_{e=0}^{h} e\ell_e. \tag{4.5}$$

For the empty minor all the $k_h(V_j)$ are zero, so it is included. Summing in h and using (4.3) gives

$$\sum_{i,j} \dim(U_i \cap V_j) \le \sum_{e=0}^{m-1} e(m-e)\ell_e \le C_m \sum_e \ell_e = C_m k_{m-1}(W) \le C_m N.$$

□

Actual disjointness is used at the node-rank step; a numerical near-disjointness is insufficient. No general-position assumption on the subspaces is needed. The infinite real field justifies the polynomial identity step.

Corollary 4.2 (Ordered rank sum)**.** *If $X_i, Y_i \in \mathbb{R}^{d\times d}$ and every $X_i - Y_i$ is invertible,*

$$\sum_{i\ne j} \operatorname{rank}(X_i - Y_j) \ge 2h(m)d. \tag{4.6}$$

Proof. Use graph subspaces $U_i = \{(v, X_i v)\}$, $V_j = \{(v, Y_j v)\}$ in $W = \mathbb{R}^d \oplus \mathbb{R}^d$. Their intersections have dimension $d - \operatorname{rank}(X_i - Y_j)$. Apply (4.1) with $N = 2d$ and remove the m diagonal ranks d. The coefficient is $m^2 - m - 2\lfloor m^2/4 \rfloor = 2h(m)$, at both $m = 2r$ and $m = 2r+1$. □

The subspace constant is sharp: split indices into balanced groups and use $(U_i, V_i) = (P, Q)$ and (Q, P) respectively for $W = P \oplus Q$. Cross-group intersections total $C_m N$. This does not say every sharp algebraic configuration is a drawing.

5 The Odd-Spoke Detector and Ram Panels

Let $n = 2s+1$, $s \ge 1$, $d = s^2$, and choose distinct real nodes $x_1, \dots, x_n$. The physical polynomial space is

$$V_s = \{P(u,v) : \deg_u P, \deg_v P \le s-1\}, \qquad \dim V_s = d,$$

with evaluation vectors $h_{pq} = (x_p^a x_q^b)_{0\le a,b<s}$. This finite tensor space, not an unrestricted function space, is the domain of the Ram panel interpretation.

Lemma 5.1 (Two-triangle unisolvency)**.** *The d evaluations at*

$$Q_s = \{(p,q) : 1 \le q < p \le s+1\} \cup \{(p,q) : s+1 < p < q \le 2s+1\}$$

uniquely determine a member of V_s.

Proof. There are $s(s+1)/2 + s(s-1)/2 = s^2$ nodes. At $s = 1$, one evaluation determines a constant. Suppose $s > 1$ and P vanishes there. The s points with $p = s+1$ imply $P(x_{s+1}, v) = 0$ identically. Thus $P = (u - x_{s+1})P_1$ with $\deg_u P_1 \le s-2$. At the remaining $s-1$ points in column $q = 1$, $P_1(u, x_1)$ vanishes, giving

$$P = (u - x_{s+1})(v - x_1)R, \qquad \deg_u R, \deg_v R \le s-2.$$

The remaining lower triangle has $2 \le q < p \le s$ and the upper triangle $s+2 \le p < q \le 2s+1$; neither factor vanishes there. On the tuple $(x_{s+2}, \dots, x_{2s+1}, x_2, \dots, x_s)$, the polynomial $R(v,u)$ vanishes on exactly Q_{s-1}. The coordinate swap exchanges the two triangles. Induction gives $R = 0$ and hence $P = 0$. □

Theorem 5.2 (M309 detector)**.** *Set*

$$F(t) = \prod_{p=1}^{n} (t - x_p), \qquad B_{pq} = \frac{x_p - x_q}{F'(x_p)F'(x_q)} h_{pq} h_{pq}^T, \tag{5.1}$$

$$\Phi(H) = \sum_{p,q} H(p,q) B_{pq}. \tag{5.2}$$

Then $\Phi : \mathbb{R}^{n\times n} \to \mathbb{R}^{d\times d}$ annihilates $\mathcal{L}$ and diagonal matrices, sends E_{pq} to rank one when $p \ne q$, and sends every half-sign matrix $S_\prec$ of a strict linear order to an invertible matrix. One fixed map works for every order.

Proof. All denominators are nonzero. The constant coordinate of h_{pq} is one, so B_{pq} has rank one for $p \ne q$ and $B_{pp} = 0$. Lagrange interpolation, followed by comparison of the coefficient of t^{n-1}, gives

$$\sum_p \frac{A(x_p)}{F'(x_p)} = 0 \qquad (\deg A \le n-2).$$

Indeed $A(t) = \sum_p A(x_p)F(t)/((t-x_p)F'(x_p))$, because the two sides agree at n distinct nodes and have degree at most $n-1$. An entry of B_{pq} has numerator $(x_p - x_q)x_p^{a+a'}x_q^{b+b'}$, of degree at most $2s-1 = n-2$ separately in either variable. Thus $\sum_q B_{pq} = \sum_p B_{pq} = 0$, annihilating $r_p + c_q$.

For the natural order let $H(p,q) = \mathbf{1}_{\{q<p\}}$. Then $S = \frac{1}{2}\mathbf{1}\mathbf{1}^T - H - \frac{1}{2}I$ and $\Phi(S) = -\Phi(H)$. Subtract one from rows $p > s+1$ to get H', leaving, off the diagonal, precisely Q_s with value $+1$ on the lower and -1 on the upper triangle. The subtraction is additive. Let A_Q have columns h_{pq} indexed by Q_s, and let

$$D_Q = \operatorname{diag}\left(H'(p,q) \frac{x_p - x_q}{F'(x_p)F'(x_q)} \right)_{(p,q)\in Q_s}.$$

Therefore

$$\Phi(S) = -A_Q D_Q A_Q^T, \quad \det \Phi(S) = (-1)^d (\det A_Q)^2 \prod_{Q_s} (D_Q)_{pq,pq} \ne 0. \tag{5.3}$$

Lemma 5.1 proves invertibility of A_Q, and all diagonal factors are nonzero. For $\pi(1) \prec \cdots \prec \pi(n)$ apply the same argument to $(x_{\pi(1)}, \dots, x_{\pi(n)})$. This reindexes the fixed sum (5.2), leaving F and Φ unchanged. The lemma holds for arbitrary distinct nodes, without requiring numerical order. □

The detector need not be positive definite; nonzero determinant is the certificate. Rank subadditivity separately yields

$$\operatorname{rank} \Phi(H) \le \#\{(p,q) : p \ne q,\ H(p,q) \ne 0\}. \tag{5.4}$$

Small floating singular values cannot replace the exact factorization.

5.1 Compatible Ram coordinates

The actual foundations passage [11] defines a finite bank by $m = Ec$, where c are coefficients in a declared physical space and $E_{ij} = \ell_i(\phi_j)$. Invertibility, or a specified left inverse on the range for a redundant bank, is essential. On a nondegenerate interval with integrable weight $w > 0$ almost everywhere, polynomial moments are unisolvent: their Gram quadratic form is $\int wp^2 > 0$ for every nonzero polynomial. Tensor products give the analogous exact representation on a rectangle. There is no Taylor remainder for members of the fixed polynomial class V_s.

Proposition 5.3 (Detector transfer with redundant panels)**.** *For full-column-rank $T \in \mathbb{R}^{r\times d}$, replacing h_{pq} by Th_{pq} gives*

$$\Phi_T(H) = T\Phi(H)T^T, \qquad \operatorname{rank}\Phi_T(H) = \operatorname{rank}\Phi(H).$$

The annihilations and rank-one bounds persist. If $r = d$, every order image is invertible. If $r > d$, a specified left inverse L gives $L\Phi_T(H)L^T = \Phi(H)$; the r-dimensional image itself is not invertible.

Proof. Substitution gives congruence and hence $\operatorname{rank}\Phi_T \le \operatorname{rank}\Phi$. With $LT = I$, the reverse congruence gives the reverse inequality. Annihilation and rank-one statements follow immediately. □

This is the exact specialization of compatible transfer in TR250 [12]. All shared coordinates require consistent maps; unrelated bases for different blocks cannot be inserted directly in $R_{ik} = X_i - Y_k$. A positive physical Gram matrix does not imply positivity of the detector. TR190 [13] supplies graph-operator language, not the interpolation or topological lemmas. Its generic finite-neighborhood triangular-inverse assertion is not used. Core application 50 [14] distinguishes cohomology transfer from preserving selected observables. Even preserving the graph cycle space does not prove plane-drawing realizability; J must still arise from the common drawing.

6 The Global Deduction with Its Interface Exposed

Theorem 6.1 (Conditional reconstruction of the source conclusion)**.** *Assume G. Then* $\operatorname{cr}(K_{m,n}) = Z(m,n)$ *for every positive m, n.*

Proof. The upper bound and zero-factor cases have been proved. Suppose $m \ge 3$, $n = 2s+1 \ge 3$ and normalize a minimum drawing. Form its common J, fix one detector-node tuple, and set

$$R_{ik} = \Phi(J^{ik}), \qquad X_i = R_{i1}, \qquad Y_k = R_{11} - R_{1k}.$$

Proposition 3.2 gives $R_{ik} = X_i - Y_k$. Each $J^{ii} = S_{\prec_i}$ has detector rank $d = s^2 = h(n)$. For $i \ne k$, Φ annihilates the possible local b_p endpoint terms at $p = q$. For $p \ne q$, a nonzero $I(ip, kq)$ requires at least one actual crossing. If c_{ik} counts all crossings between stars i, k, then (5.4) gives $\operatorname{rank} R_{ik} \le c_{ik}$ even with repeated edge crossings. Signed cancellation only weakens this bound. Corollary 4.2 therefore gives

$$2h(m)h(n) \le \sum_{i\ne k} \operatorname{rank} R_{ik} \le \sum_{i\ne k} c_{ik} = 2c(D).$$

Every crossing occurs in two ordered star pairs, explaining the factor. For even $n > 2$, delete each vertex in B. Every crossing uses two distinct B-endpoints and survives exactly $n-2$ deletions. The odd bound gives

$$(n-2)c(D) = \sum_b c(D-b) \ge nh(m)h(n-1) = (n-2)h(m)h(n).$$

Divide by $n-2$. The case $n=2$ was already handled without division, and symmetry of the parts covers all cases. □

This is a complete implication from G, not an inference from a small test suite. It is not the private pair-frequency relaxation and does not require a conjectural sharp value of that relaxation.

6.1 Step-level disposition

Source step	Required invariant	Disposition
`normalization.tex`	Continuous arcs, ordinary point count, no new improper crossings	Full reading and scoped manual audit; construction exposed in G.
`intersections.tex`	Common drawing, signs, shared-edge averaging, cycle pairing	Source construction reconstructed; local chord proof and exact common-drawing checks.
`lem:additive`	All rectangular differences, also degenerate indices	Complete algebraic proof.
`subspaces.tex`	Distinct nodes, true disjointness, nonzero minor, degree and vanishing multiplicity	Complete proof and independent finite filtration tests.
`detector.tex`	Dimension s^2, one map for all orders, no positivity assumption	Complete proof and exact determinants.
`conclusion.tex`	Full diagonal rank, actual crossing counts, ordered factor two	Complete deduction from G.
Even deletion	Two distinct used endpoints; $n>2$ before division	Exact pointwise counting and parity check.

7 The Private Rotation-Frequency Route

For fixed $m \geq 3$, let Ω_m be the $(m-1)!$ cyclic orders of the labeled A-vertices. A drawing gives an order $\gamma(b)$ at each $b \in B$. Define $Q_{\sigma\tau}$ by the minimum crossing count between two stars in an actual $K_{m,2}$ drawing with these orders. The classical method uses $Q_{\sigma\sigma} = h(m)$ [3, 6]. This is an imported rotation fact, not a consequence of a Ram change of variables. Do not confuse Q with the detector pattern Q_s. Let $c_\sigma = \#\{b : \gamma(b) = \sigma\}$ and $x = c/n$; define

$$q_m = \min\{x^T Q x : x \geq 0,\ \mathbf{1}^T x = 1\}.$$

Every drawing supplies an integral count vector with $\mathbf{1}^T c = n$. Conversely, a count vector and independently optimal two-star drawings need not be realizable simultaneously.

Proposition 7.1 (Frequency and copositive certificates)**.** *For every normalized drawing,*

$$2c(D) \geq c^T Q c - h(m)n, \qquad \mathrm{cr}(K_{m,n}) \geq \left\lceil \frac{q_m n^2 - h(m)n}{2} \right\rceil. \tag{7.1}$$

Moreover $q_m \geq \alpha$ exactly when $Q - \alpha \mathbf{1}\mathbf{1}^T$ is copositive.

Proof. Each pair b, b' contributes at least $Q_{\gamma(b),\gamma(b')}$. The quadratic expansion counts both orders and includes n artificial self-star terms $h(m)$, while every actual crossing belongs to one unordered pair. This proves the first inequality. Normalize by n, then use integrality of the crossing count. On the simplex $x^T\mathbf{1}\mathbf{1}^T x = 1$. Every nonzero $y \geq 0$ equals $(\mathbf{1}^T y)x$ for such an x, and quadratic forms scale by $(\mathbf{1}^T y)^2$, proving the equivalence. □

An exact identity $Q - \alpha\mathbf{1}\mathbf{1}^T = S + N$, $S \succeq 0$ and entrywise $N \geq 0$, is sufficient. Rational LDL^T or an exact sum-of-squares identity certifies S; floating solver output alone does not. For any finite permutation group preserving Q, averaging S, N preserves the identity and both cones by convexity. Thus invariant certificates lose no strength at that relaxation level. Arbitrary spectral changes do not preserve entrywise nonnegativity.

7.1 Domains in Ram coordinates

If $c = Ry$, the form becomes $y^T R^T Q R y$, but feasibility becomes

$$Ry \geq 0, \qquad \mathbf{1}^T R y = n, \qquad Ry \in \mathbb{Z}^{|\Omega_m|}.$$

Replacing this with the usual positive integer simplex in y changes the problem. Copositivity transfers with its cone, and integrality with its lattice. A global reversal acts on all rotations; independently reversing each type is not an unrestricted symmetry.

7.2 Finite-size correction and exact residuals

Proposition 7.2 (Ceiling-sensitive odd correction)**.** *Suppose $q_m \geq h/2$, where $h = h(m)$. For even n,* (7.1) *reaches $Z(m,n)$. For odd n it gives exactly*

$$\left\lceil h(n^2 - 2n)/4 \right\rceil = Z(m,n) - \lfloor h/4 \rfloor. \tag{7.2}$$

Rounding alone removes this gap if $h \leq 3$, in particular $m = 3, 4$.

Proof. For even n, $h(n) = (n^2 - 2n)/4$. For odd n, $h(n) = (n^2 - 2n + 1)/4$. The expression inside the ceiling is the integer $Z(m,n)$ minus $h/4$, and $\lceil Z - a \rceil = Z - \lfloor a \rfloor$ for integral Z. □

The ideal coefficient remains a hypothesis. Before rounding the odd gap is $h/4$, exactly as in the private report; this arithmetic refinement does not establish a universal solution.

Suppose instead an exact identity is available:

$$Q - \alpha\mathbf{1}\mathbf{1}^T = A^T D A + N, \qquad D = \operatorname{diag}(d_\ell),\ d_\ell \geq 0, \quad N \geq 0.$$

If every admissible integer c satisfies $\sum_\ell d_\ell (Ac)_\ell^2 + c^T N c \geq \rho(n)$, substitution in (7.1) proves

$$\operatorname{cr}(K_{m,n}) \geq \left\lceil \frac{\alpha n^2 - hn + \rho(n)}{2} \right\rceil.$$

An integer row of A all of whose entries are odd evaluates to an integer congruent to n modulo two, so it contributes at least its coefficient when n is odd. For rational rows clear and retain the denominator. Coupled row residues can strengthen separate lower bounds.

For $m = 3$ there are two orders and $Q = I_2$. Opposite orders admit a planar two-star drawing. Identical orders cannot: three internally disjoint paths between two centers have reversed endpoint cyclic orders in a plane embedding. One crossing realizes the other order. With $u = (1, -1)^T$, $I_2 = \frac{1}{2}\mathbf{1}\mathbf{1}^T + \frac{1}{2}uu^T$, hence

$$2c(D) \geq \tfrac{1}{2}n^2 - n + \tfrac{1}{2}(c_1 - c_2)^2, \qquad (c_1 - c_2)^2 \geq n \bmod 2.$$

Together with the upper bound this recovers the known $\operatorname{cr}(K_{3,n}) = h(n)$ including every floor term.

8 Proved Local Refinements

8.1 Subspace defect and equality restrictions

Theorem 8.1 (Nonnegative defect decomposition)**.** *With the notation of Theorem 4.1, put*

$$b_h = \sum_{e=0}^{h} e\ell_e - \sum_j k_h(V_j) \geq 0, \qquad K = \sum_{e=0}^{m-1} \ell_e \leq N.$$

Then

$$C_m N - \sum_{i,j} \dim(U_i \cap V_j) = C_m(N-K) + \sum_{e=0}^{m-1} [C_m - e(m-e)]\ell_e + \sum_{h=0}^{m-1} b_h \tag{8.1}$$

is an exact sum of nonnegative integers. For $m \geq 2$, equality forces $K = N$, every $b_h = 0$, and $\ell_e = 0$ outside the middle degrees $\{\lfloor m/2 \rfloor, \lceil m/2 \rceil\}$.

Proof. From (4.3) and the definition of b_h,

$$\sum_{i,j} \dim(U_i \cap V_j) = \sum_h \left(\sum_{e \leq h} e\ell_e - b_h \right) = \sum_e e(m-e)\ell_e - \sum_h b_h.$$

Subtract from $C_m N$, and add and subtract $C_m K$. Nonnegativity follows from (4.5), $K \leq N$, and $e(m-e) \leq C_m$. At $m \geq 2$, $C_m > 0$ and the maximum of $e(m-e)$ occurs only at those middle integers. Every term must therefore vanish in the equality case. □

This identifies dimension loss, degree-distribution loss, and multiplicity-budget loss separately. The b_h are exact filtration quantities, not unspecified numerical residual norms.

Corollary 8.2 (Crossing slack decomposition)**.** *For an odd-n common-drawing certificate satisfying G, set $r_{ik} = \operatorname{rank} R_{ik}$. Let Δ be the right side of* (8.1) *for the graph subspaces used in Corollary 4.2, with $N = 2d$. Then*

$$2(c(D) - Z(m,n)) = \sum_{i \neq k} (c_{ik} - r_{ik}) + \Delta. \tag{8.2}$$

An extremal drawing therefore has $c_{ik} = r_{ik}$ in every ordered off-diagonal pair, and all the filtration equality restrictions.

Proof. For graph subspaces the left side of (8.1) equals $\sum_{i \neq k} r_{ik} - 2h(m)d$. Add the crossing-rank slack and use $\sum_{i \neq k} c_{ik} = 2c(D)$ and $d = h(n)$. Every summand is nonnegative; equality forces each to vanish. □

These are necessary conditions, not an extremal-drawing classification. Equality cannot hide signed cancellation or rank-subadditivity loss in one star pair behind an improvement in another.

8.2 Replication-weighted rank certificates

Theorem 8.3. *If every $X_i - Y_i$ is invertible of size d, and w_i are nonnegative integers with $L = \sum_i w_i \geq 1$, then*

$$\sum_{i \neq j} w_i w_j \operatorname{rank}(X_i - Y_j) \geq d \left[L^2 - \sum_i w_i^2 - 2\lfloor L^2/4 \rfloor \right]. \tag{8.3}$$

For arbitrary real $w_i \geq 0$, the homogeneous consequence is

$$\sum_{i \neq j} w_i w_j \operatorname{rank}(X_i - Y_j) \geq d \left(L^2/2 - \sum_i w_i^2 \right). \tag{8.4}$$

Proof. Make w_i identical copies of (X_i, Y_i) and apply Corollary 4.2 to the L indices. Distinct copies of the same pair contribute d, so its ordered sum is

$$\sum_{i \neq j} w_i w_j \operatorname{rank}(X_i - Y_j) + d \sum_i w_i(w_i - 1).$$

Subtract the latter term from $2h(L)d$, using $2h(L) = L^2 - L - 2\lfloor L^2/4 \rfloor$, to obtain (8.3). For rational weights clear a common denominator, divide the integer inequality by its square, and discard the nonnegative parity correction. Approximate nonnegative real weights by rational ones; both finite quadratic expressions are continuous. Zero total weight is immediate. □

For an odd-n drawing satisfying G, this implies

$$\sum_{i<j} w_i w_j c_{ij} \geq \frac{h(n)}{2} \left[L^2 - \sum_i w_i^2 - 2\lfloor L^2/4 \rfloor \right]$$

for integral weights; take the maximum with zero if the right side is negative. It recovers the unweighted bound at $w_i = 1$. It is an algebraic weighted extension under G, not a claimed exact formula for weighted crossing minimization.

8.3 Weighted frequency counts and error recovery

For nonnegative masses t_b on B, let $z_\sigma = \sum_{\gamma(b)=\sigma} t_b$ and $T = \sum_b t_b$. The pair expansion from Proposition 7.1 proves

$$2 \operatorname{cross}_t(D) \geq z^T Q z - h(m) \sum_b t_b^2 \geq \alpha T^2 - h(m) \sum_b t_b^2$$

if $Q - \alpha \mathbf{1}\mathbf{1}^T$ is copositive. The self-star term is $h(m) \sum_b t_b^2$, not $h(m)T$. This is the private report's weighted result, distinct from the replication extension above.

Suppose a recovered identity is $Q - \alpha \mathbf{1}\mathbf{1}^T = S + N + E$, with $S \succeq -\eta I$, $N \geq 0$ entrywise, $E_{ij} \geq -\varepsilon$, and certified $\eta, \varepsilon \geq 0$. For a simplex vector, $\|x\|_2^2 \leq 1$ and $x^T E x \geq -\varepsilon$. Thus

$$q_m \geq \alpha - \eta - \varepsilon, \qquad \operatorname{cr}(K_{m,n}) \geq \left\lceil \frac{(\alpha - \eta - \varepsilon)n^2 - h(m)n}{2} \right\rceil.$$

This is a complete proof of the error budget, with its n^2 amplification. Uncertified floating residuals are not valid bounds for η, ε. A nonorthogonal block basis requires transport of the physical norm before importing spectral tolerances.

9 Deletion, Parity, and Higher Overlaps

Proposition 9.1 (Exact one-part defect identity)**.** *For a normalized drawing put* $\delta(D) = c(D) - h(m)h(n)$*. For* $n > 2$*,*

$$\sum_{b\in B} c(D-b) = (n-2)c(D), \tag{9.1}$$

$$(n-2)\delta(D) = \sum_{b} \delta(D-b) - h(m)\varepsilon_n, \tag{9.2}$$

$$\varepsilon_n = \begin{cases} 0 & n \text{ even}, \\ (n-1)/2 & n \text{ odd}. \end{cases}$$

Proof. Each crossing has exactly two distinct B-endpoints and survives $n-2$ deletions. Substitute the defects in this pointwise identity. The remaining scalar identity is $(n-2)h(n) - nh(n-1) = \varepsilon_n$. At $n = 2r$ it is zero; at $n = 2r+1$ it is $(2r-1)r^2 - (2r+1)r(r-1) = r$. □

A smallest counterexample, if any, has both parts odd: a negative defect at an even part forces a smaller negative-defect deletion. At an odd part, mere nonnegativity of deletion defects does not suffice; their sum must reach $h(m)(n-1)/2$. The source detector addresses the odd case directly and avoids an invalid induction from even cases.

The two-part overlap identity is, for $2 \le a \le m$, $2 \le b \le n$,

$$\sum_{\substack{A'\subseteq A,\ |A'|=a \\ B'\subseteq B,\ |B'|=b}} c(D[A',B']) = \binom{m-2}{a-2}\binom{n-2}{b-2} c(D). \tag{9.3}$$

Each crossing specifies two vertices in each part; the remaining vertices are freely chosen. This proves the identity without assuming pairwise optimal subdrawings. If every such drawing has a certified lower bound $L_{a,b}$, division gives

$$c(D) \ge \frac{\binom{m}{a}\binom{n}{b}}{\binom{m-2}{a-2}\binom{n-2}{b-2}} L_{a,b}.$$

This classical sampling principle propagates genuine higher-order information. It cannot manufacture common-drawing compatibility from independent two-star minima.

10 Exact Checks and Available Lean Artifacts

10.1 Independent author implementation

The self-contained checker is `_checks/m309_exact_audit.py`; its output is `_checks/m309_exact_results.json`, under this report's directory. All evidence arithmetic uses Python integers and `fractions.Fraction`, including custom rational elimination for ranks and determinants. No floating diagnostic is a certificate.

s	$n = 2s+1$	Orders checked	$\det A_{Q_s}$ at $x_p = p$
1	3	all 6	1
2	5	all 120	−4
3	7	3 selected	−14400
4	9	3 selected	−264290829336576

All row and column sums of B_{pq} vanish exactly; every $B_{pp} = 0$. The natural-order factorization is verified, and each tested order has nonzero determinant. At $n = 5$ the natural-order determinant is $-1/1492992$, explicitly confirming that this is not a positive-definite detector. The larger selected orders are natural, reverse, and even-index-first, not an exhaustive census.

Six actual axis drawings use balanced signed powers of two on the vertical axis and signed powers of three on the horizontal axis. Exact segment equations and determinant signs identify crossings; rational intersection coordinates exclude triple concurrence. Half-plane and cross-product comparisons determine spoke orders, without trigonometry. This one geometry supplies J, every cycle, and every detector block.

(m, n)	$c(D)$	Cycle dimension	Diagonal rank	Ordered rank sum
$(3, 3)$	1	4	1	2
$(4, 3)$	2	6	1	4
$(3, 5)$	4	8	4	8
$(4, 5)$	8	12	4	16
$(5, 5)$	16	16	4	32
$(3, 7)$	9	12	9	18

In each drawing the checks prove $C^T JC = 0$, the additive block relation, diagonal full rank, off-diagonal rank budgets, and $c(D) = Z(m, n)$. All $K_{3,n}$ count splits through $n = 80$ and the deletion and rounding identities through $m, n = 80$ pass in their declared domains. The weighted inequality passes for all 216 scalar three-index families with entries in $\{0, 1, 2\}$ and unequal diagonal pairs, at weights $(1, 1, 1), (1, 2, 3), (2, 0, 1)$. All ten manuscript files have matching public-pin bytes and recorded SHA-256 values.

10.2 Independent algebra audit

The separately authored note and implementation were both read:

```
C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_audit/M309-independent-alg
ebra-audit.txt
C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_audit/M309-independent-alg
ebra-check.js
```

The note reports no concrete algebra error. Its JavaScript checker uses BigInt elimination and Bareiss determinants, scaling barycentric matrices by an exact common denominator squared. An independent local replay and our own replay both passed. Its coverage is:

- All 6, 120, and 5040 orders at $n = 3, 5, 7$ respectively; 30 orders each at $n = 9, 11$ with nonuniform, unsorted signed nodes. It checks annihilation, coordinate ranks, interpolation determinants, and equality of direct and factored detector images.
- 100 deterministic sampled disjoint-partner subspace families, 368 filtration stages, evaluation identities, node-rank loss, degree/multiplicity budgets, and final intersection sums.
- Rank constants and sharpness examples for $m = 1, \ldots, 30$, and even deletion identities for $n = 4, 6, \ldots, 30$.
- 36 generic straight-line drawings, with $m = 3, 4, 5, 6$ and $n = 3, 5, 7$, testing 9072 ordered cycle pairings and the common-drawing block and rank interfaces.

The separate implementation reduces the risk of a single coding mistake. These finite checks are corroboration, not quantification over arbitrary nodes, subspaces, or repeated-crossing topological configurations. No new universal crossing constant, SDP optimization, or exhaustive drawing census was computed.

10.3 Bounded formalization inspection

Independent local inspection of the pinned formalization catalogue lists `OAI.Zarankiewicz.mainTarget_proof` in `OAI/Combinatorics/Crossing/Main.lean`. We read that complete solution file, the complete challenge file, and the comparator JSON from the pinned local acquisition. The challenge defines injective vertices and simple continuous path edges avoiding all interior vertices. Proper crossings use local open partial homeomorphisms. Admissibility requires finitely many crossing points and no triple intersection, but does not assume no adjacent crossings or at most one crossing per edge pair. Its target includes both an axis-formula drawing and the lower bound for every admissible drawing at positive m, n.

The solution invokes `nonempty_balancedAxisRealization`, its `crossingCount_eq`, and imported `crossing_lower_bound` to prove the named target. It then derives attainment and the equality

$$\texttt{ordinaryCrossingNumber}\ m\ n = \texttt{axisPairs}\ m \cdot \texttt{axisPairs}\ n.$$

The JSON permits `propext`, `Quot.sound`, and `Classical.choice`, with `enable_nanoda:false`. The challenge's `sorry` is an intentional specification placeholder, not evidence of a hole in the solution. Conversely, the top-level invocation does not validate its transitive imports. Lean, the comparator, and an independent kernel were not replayed by this report. Formalization is available; calling it unformalized would be inaccurate.

11 Benefits, Limits, and Future Work

Benefits. Polynomial Ram coordinates retain the exact detector and its common drawing constraints. Frequency coordinates expose the positive integer simplex and enable rational copositive certificates. The defect identity separates dimension, degree, and multiplicity losses; crossing slack then separates them from signed cancellation and rank subadditivity. Integer replication supplies weighted bounds with explicit parity, without guessing a weighted topological formula.

Limitations. No invertible coordinate map or finite Gram matrix establishes G or realizes arbitrary pair data. The rotation state count is factorial; copositivity is hard; SDP bounds can be strict. Detector dimension is s^2 and rational denominators can grow rapidly. Equality restrictions are necessary, not a full classification. The positive manual audits, finite checks, and named Lean theorem have distinct scopes. This is not a publication-priority determination or a substitute for dependency and comparator replay.

Concrete future work. Replay the pinned Lean comparator and inspect all dependencies at the geometric interface. Formalize the shared-edge shifts and seam transversality. Compute filtered profiles ℓ_e, b_h for varied extremal drawings to assess how restrictive (8.1) is. Construct sparse exact evaluation bases that reduce coefficient growth while retaining congruence certificates. Couple integer residues to genuine three-star and larger overlap constraints. Higher floating precision alone accomplishes none of these logical tasks.

Conclusion. The direct private report is a core match to M309, not a peripheral analogy: it preserves ordinary crossings, finite parity, rotation states, and realizability. The pinned manuscript supplies a different signed-intersection and rank route. Fully proved local algebra supports its transparent conditional reconstruction and the new defect and weighted refinements. The September 6 baseline, September 23 claim, exact finite checks, manual geometric review, and unreplayed formal artifacts remain explicitly separated.

A Exact Paths and Reading Depth

Maps were navigation aids, not substitutes for actual passages. No full-corpus reading is claimed. Private material stayed local.

Catalogue and manuscript.

`C:/Users/mural/OneDrive/Documents/OpenAI-722-math/00-Applicability-Catalogue/applicability.json`
`C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_sources/openai-math/preprints/The-crossing-number-of-complete-bipartite-graphs-September-23-2026/build`

Row 309 read in full. Complete reading of `paper.tex`, `references.bib`, `sections/introduction.tex`, `sections/subspaces.tex`, `sections/intersections.tex`, `sections/detector.tex`, `sections/conclusion.tex`, `sections/normalization.tex`, `figures/chord-pairing.tex`, and `figures/triangular-grid.tex`. Hashes and exact remote-pin comparisons are in the local results JSON.

Direct private source, fully read.

`C:/Users/mural/OneDrive/Documents/MyDocuments2026/res-rt/res-math/TR-item-071-astra.tex`

Read from document class through bibliography and end-document, including corrected attribution, finite-size arithmetic, deletion, weighted counts, robust recovery, and corpus limitations.

Navigation maps, selected routing entries.

`C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_audit/corpus-core-map.txt`
`C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_audit/corpus-foundations-map.txt`
`C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_audit/corpus-applications-map.txt`

Coverage warnings and applicable entries were inspected, notably core E5/E14, foundations P1, and application C50. Their broad inventory is not represented as our independent reading.

Actual additional private passages.

1. `C:/Users/mural/OneDrive/Documents/RamFoundationTRs/TR-rev-1-old-tr250/TR-rev-1-old-tr250.tex`. Read the finite-representation subsection, `eq:moments`, and the concrete polynomial construction, including unisolvency, range constraints, and quadrature limitations.

2. C:/Users/mural/OneDrive/Documents/MyDocuments2026/res-rt/TechReps/FinalTechReps/TRs/TR250-Compatible-Variational-Ram-Panels/TR250-Compatible-Variational-Ram-Panels.tex. Read thm:transfer and its proof, common-degree and metric cautions, curved-panel discussion, and bounded-projection limitations.

3. C:/Users/mural/OneDrive/Documents/MyDocuments2026/res-rt/TechReps/FinalTechReps/TRs/TR190-GGRT/TR190-GGRT.tex. Read def:atom, def:forward, def:inverse and sec:scope. No generic triangularity or speedup is imported.

4. C:/Users/mural/OneDrive/Documents/RamPatsCore50-TRs/TR-core-pat-50/TR-core-pat-50.tex. Read transfer identities, thm:homotopy and proof, periods versus observables, approximate-homotopy limitations, composition, and prop:separating with proof.

TR188, TR191, TR248, and TR249 occur in the fully read direct report, but their original full reports were not read for this task; none is independently credited as a proved input here. Other listed corpus roots were not searched exhaustively.

Formal artifacts, fully read at the named-file level.

```
C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_sources/lean-batch5/OAI/Combinatorics/Crossing/Main.lean
C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_sources/lean-batch5/ComparatorChallenges/BipartiteCrossing.lean
C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_sources/lean-batch5/ComparatorChallenges/BipartiteCrossing.json
```

The pinned catalogue observation was independently checked locally. Transitive imports and comparator execution remain outside this reading boundary.

Author evidence and replay files.

```
C:/Users/mural/OneDrive/Documents/OpenAI-722-math/M309-Bipartite-Crossing-Numbers/_checks/m309_exact_audit.py
C:/Users/mural/OneDrive/Documents/OpenAI-722-math/M309-Bipartite-Crossing-Numbers/_checks/m309_exact_results.json
C:/Users/mural/OneDrive/Documents/OpenAI-722-math/M309-Bipartite-Crossing-Numbers/_checks/M309-author-evidence.txt
```

The separate audit note and JavaScript paths are listed in Section 10.1's independent-audit discussion; both were read in full. Original files, catalogue, queue, and progress records were not edited.

References

[1] OpenAI, *The crossing number of complete bipartite graphs*, September 23, 2026, commit adc7f1241b42e322a6451854ab7e4b4c146bf78a. https://github.com/openai/math/tree/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/The-crossing-number-of-complete-bipartite-graphs-September-23-2026. Complete local mathematical sources read.

[2] M. Subbarao, *Ram Calculus for Turán's Brick-Factory Problem: Rotation Systems, Copositive Certificates, and Symmetry-Reduced Bounds*, private report, September 6, 2026. Full path and reading depth in Appendix A.

[3] D. Brosch and S. C. Polak, New lower bounds on crossing numbers of $K_{m,n}$ from semidefinite programming, *Mathematical Programming* 207 (2024), 693–715. https://doi.org/10.1007/s10107-023-02028-1. Primary text: https://ir.cwi.nl/pub/33564/33564.pdf. Abstract, Sections 1–2, and symmetry discussion consulted; its full numerical computation was not replayed.

[4] S. C. Polak, author publication record, https://sites.google.com/site/svenpolak/home. Bibliographic metadata checked October 7, 2026.

[5] J. Balogh, B. Lidický, S. Norin, F. Pfender, G. Salazar, and S. Spiro, Crossing numbers of complete bipartite graphs, *Procedia Computer Science* 223 (2023), 78–87. https://doi.org/10.1016/j.procs.2023.08.216. Primary institutional record and abstract checked: https://experts.illinois.edu/en/publications/crossing-numbers-of-complete-bipartite-graphs/.

[6] D. R. Woodall, Cyclic-order graphs and Zarankiewicz's crossing-number conjecture, *Journal of Graph Theory* 17 (1993), 657–671. https://doi.org/10.1002/jgt.3190170602. Historical rotation input identified in source and primary modern literature; not reread in full here.

[7] D. J. Kleitman, The crossing number of $K_{5,n}$, *Journal of Combinatorial Theory* 9 (1970), 315–323. https://doi.org/10.1016/S0021-9800(70)80087-4. Historical deletion attribution; full article not audited here.

[8] J. Pach and G. Tóth, Which crossing number is it anyway?, *Journal of Combinatorial Theory B* 80 (2000), 225–246. https://doi.org/10.1006/jctb.2000.1978. Source bibliography context for drawing conventions.

[9] R. Fulek, M. J. Pelsmajer, M. Schaefer, and D. Štefankovič, Adjacent crossings do matter, *Journal of Graph Algorithms and Applications* 16 (2012), 759–782. https://doi.org/10.7155/jgaa.00266. Source bibliography context; not a substitute for normalization.

[10] J.-P. Berrut and L. N. Trefethen, Barycentric Lagrange interpolation, *SIAM Review* 46 (2004), 501–517. https://doi.org/10.1137/S0036144502417715. Classical background; the used identity is proved here.

[11] M. Subbarao, private revised foundational report TR-rev-1-old-tr250, finite moment representations and polynomial construction; path and selected reading in Appendix A.

[12] M. Subbarao, *Compatible Variational Ram Panels*, TR250, private technical report, compatible coordinate transfer; path and selected reading in Appendix A.

[13] M. Subbarao, *The Graph-Global Ram Transform*, TR190, private technical report, definitions and limitations; path and selected reading in Appendix A.

[14] M. Subbarao, *Period Preserving Transfer of Physical Field States During Mesh Adaptation*, private core application 50; finite homotopy and separating periods, path in Appendix A.